



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
12-07-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

PAPER-2

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

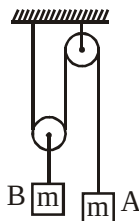
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. In the figure shown assume ideal string and pulley, the acceleration of the blocks A and B are:-



(A) $a_A = \frac{g}{5}, a_B = \frac{2g}{5}$ (B) $a_A = \frac{2g}{5}, a_B = \frac{g}{5}$ (C) $a_A = \frac{g}{2}, a_B = \frac{g}{4}$ (D) $a_A = \frac{g}{4}, a_B = \frac{g}{2}$

Space for Rough Work

2. The length, breadth and height of a cuboid depend on time as

$$L = \sin t$$

$$B = t^2$$

$$C = t$$

The rate of change of volume with time at $t = \frac{\pi}{2}$ is $\frac{k\pi^2}{4}$. Value of k is.

(A) 4

(B) 5

(C) 3

(D) 2

3. A tennis ball launcher is placed on the floor at the front end of a train that has a uniform forward acceleration of 2 m/s^2 . The launcher projects a ball at an initial speed of 20 m/s with respect to the train toward its rear. The ball achieves a maximum height of 10.0 m . Ignore air resistance. Find how far (in m) from the front end of the train the ball lands inside the train.

(A) 24

(B) 48

(C) 45

(D) 96

4. At an instant lift is moving with 10 m/s upward and with acceleration 5 m/s^2 upward. At the same instant, a coin is dropped from 7.5 m above the floor of lift, find the distance travelled by coin in ground frame before it strikes the floor :-

(A) 5 m (up)

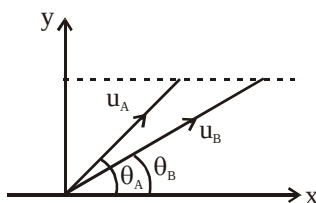
(B) 5 m (down)

(C) 10 m (up)

(D) 10 m (down)

Space for Rough Work

5. Two particles A and B are projected from the same point in different directions in such a manner that vertical component of their initial velocity are same then ratio of their horizontal range will be equal to :-



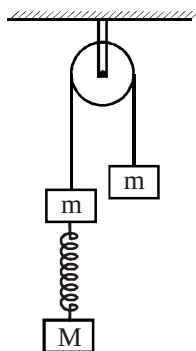
- (A) $\frac{R_A}{R_B} = \frac{u_A}{u_B}$ (B) $\frac{R_A}{R_B} = \frac{\tan \theta_B}{\tan \theta_A}$ (C) $\frac{R_A}{R_B} = \frac{\tan \theta_A}{\tan \theta_B}$ (D) $\frac{R_A}{R_B} = \frac{\sin \theta_A}{\sin \theta_B}$
6. A cannon ball has a range R on a horizontal plane, such that the corresponding possible maximum heights reached are H_1 and H_2 . Then, the correct expression for R is :-
- (A) $\frac{(H_1 + H_2)}{2}$ (B) $(H_1 H_2)^{1/2}$ (C) $2(H_1 H_2)^{1/2}$ (D) $4(H_1 H_2)^{1/2}$
7. A police jeep is chasing with, velocity of 45 km/h a thief in another jeep moving velocity 153km/h. Police fires a bullet with muzzle velocity of 180 m/s. The velocity it will strike car of the thief is
- (A) 150 m/s (B) 27 m/s (C) 450 m/s (D) 250 m/s

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
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- For each question, choose the correct option(s) to answer the question.
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
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Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

8. The system shown in figure is released from rest with spring at natural length. The spring starts extending as soon as it is released **if** (Neglect friction and masses of pulley, string and spring)



(A) $M > m$

(B) $M > 2m$

(C) $M > m/2$

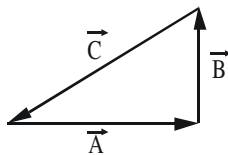
(D) $M > m/3$

Space for Rough Work

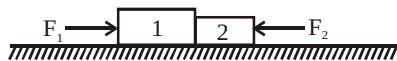
9. Consider the curve representing function $y = x^2 + x + 1$ on a Cartesian x-y plane. Which of the following statements are true?
 (A) It crosses the y-axis at point (0, 1).
 (B) Slope of tangent to the curve is unity at point (0, 0).
 (C) Graph between dy/dx and x is a straight line.
 (D) Graph between dy/dx and x crosses the y-axis at point (0, 1).
10. A particle of mass 0.3 kg subject to a force $F = -kx$ with $k = 15 \text{ N/m}$. What will be its initial acceleration if it is released from a point 20 cm away from the origin?
 (A) 15 m/s^2 (B) 3 m/s^2 (C) 10 m/s^2 (D) 5 m/s^2
11. Displacement of a particle is given as :
 $x = A e^{-kt} \sin (Bt + C)$
 which of the following is/are **CORRECT** :-
 (A) $\left(\frac{B}{k}\right)$ is dimensionless. (B) $(B + k)$ may be a meaningful term.
 (C) Dimension of $\frac{A}{k^2}$ is $[LT^{-2}]$ (D) $[A] = [C]$

Space for Rough Work

12. Which expression is/are **CORRECT** concerning the vectors shown in the sketch?



- (A) $\vec{C} = \vec{A} + \vec{B}$ (B) $\vec{C} + \vec{A} = -\vec{B}$ (C) $\vec{A} + \vec{B} + \vec{C} = 0$ (D) $|\vec{C}| = |\vec{A} + \vec{B}|$
13. A displacement vector of magnitude 10 m has its initial point at (4, 3)m. The y-component of this vector has a magnitude of 6 m. The co-ordinates of the final point of the vector may be
 (A) (-4, 9) m (B) (-4, -3) m (C) (12, 9) m (D) (12, -3) m
14. Two blocks 1 and 2, on a frictionless table, are pushed from the left by a horizontal force $\vec{F}_1$, and on the right by a horizontal force $\vec{F}_2$ as shown below. The magnitudes of the pushing forces satisfy the inequality $|\vec{F}_1| > |\vec{F}_2|$. Which of the following statements is true about the magnitude N of the contact force between the two blocks ?



- (A) $N < |\vec{F}_1|$ (B) $N > |\vec{F}_1|$ (C) $N < |\vec{F}_2|$ (D) $N > |\vec{F}_2|$

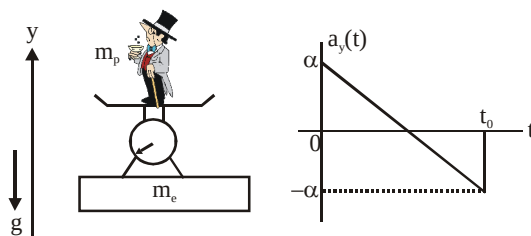
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

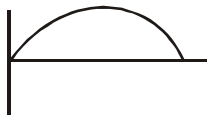
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Paragraph for Question no. 15 and 16

A person of mass m_p stands on a scale in an elevator of mass m_e . The scale reads the magnitude of the force F exerted on it from above in a downward direction. Starting at rest at $t = 0$ the elevator moves upward, coming to rest again at time $t = t_0$. The downward acceleration of gravity is g . The acceleration of the elevator during this period is shown graphically below.



15. What will be the reading on scale when speed of elevator is maximum :-
 (A) $m_p g$ (B) $m_e g$ (C) $(m_p - \alpha)g$ (D) $(m_p + \alpha)g$
16. Find the total distance traveled by the elevator before coming to rest

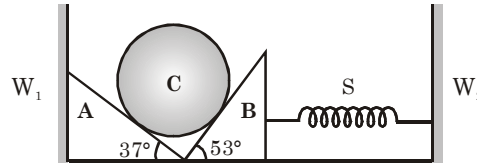


- (A) $\frac{\alpha t_0^2}{24}$ (B) $\frac{\alpha t_0^2}{6}$ (C) $\frac{\alpha t_0^2}{8}$ (D) $\frac{\alpha t_0^2}{12}$

Space for Rough Work

Paragraph for Question no. 17 and 18

In the figure is shown a system of three blocks A, B & C held in equilibrium between two walls W_1 and W_2 and spring S. Each of A, B & C weigh 100 N.



17. The force applied by wall W_1 on block A is
 (A) 40 N (B) 48 N (C) 64 N (D) 80 N
18. The force applied by spring S on block B is
 (A) 24 N (B) 96 N (C) 48 N (D) 64 N

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 21)**

- This section contains **SEVEN** questions.
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1. Which of the following general formula represent pyroxene :



2. Which of the following non metal does **NOT** form dibasic oxyacid?



3. The **CORRECT** order of boiling point is :

- (A) n-pentane < neo pentane
(B) p-nitrophenol < o-nitrophenol
(C) $\text{NH}_3 < \text{HF} < \text{H}_2\text{O}$
(D) $\text{H}_2\text{O} > \text{D}_2\text{O}$

Space for Rough Work

4. Which of the following molecular species exist ?
(A) OF_4 (B) XeF_3^+ (C) PBr_6^- (D) BF_6^{3-}
5. XeO_xF_y is non planar and nonpolar then value of x and y respectively is.
(A) 3, 0 (B) 2, 2 (C) 1, 4 (D) 3, 2
6. Which of them is shape of cationic and anionic part of solid Cl_2O_6 respectively.
(A) angular, angular (B) pyramidal, tetrahedral
(C) angular, tetrahedral (D) linear, angular
7. Which pair of species has identical bond angle.
(A) ClF_3 , BrF_3 (B) PF_3 , PCl_3 (C) CO_2 , SO_2 (D) CCl_4 , SiCl_4
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

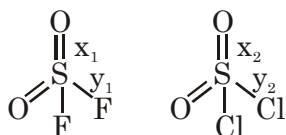
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-
8. Which of the following molecular species is/are linear.
- (A) I_3^- (B) N_3H (C) $[I(CN)_2]^-$ (D) S_3^{2-}
9. Select **CORRECT** statement/s.
- (A) Smallest anion in the periodic table is H^-
- (B) Ionic mobility of $Mg_{(aq)}^{2+}$ is greater than that of $Be_{(aq)}^{2+}$
- (C) Electronegativity of Pb^{2+} is more than that of Pb^{4+}
- (D) Second ionisation energy of Al is higher than second ionisation energy of Mg
10. Which of the following orbital(s) is/are commonly used by central atom in hybridisation of XeF_4 and SF_4 .
- (A) p_x (B) p_z (C) $d_{x^2-y^2}$ (D) d_{z^2}

Space for Rough Work

11. Which of the following property(ies) is/are associated with an element having $Z = 80$.

- (A) Group no. = 12
- (B) Period no. = 6
- (C) Transition metal
- (D) Exist as liquid at 25°C

12. **CORRECT** order/s for following structure is/are :



- (A) $x_1 > x_2$
- (B) $x_1 < x_2$
- (C) $y_1 > y_2$
- (D) $y_1 < y_2$

13. Select the **CORRECT** order/s of lattice energy?

- (A) $\text{CaO} < \text{BaO}$
- (B) $\text{MgO} < \text{Al}_2\text{O}_3$
- (C) $\text{NaF} < \text{CsF}$
- (D) $\text{CaF}_2 < \text{CaO}$

14. Consider Ga ($Z = 31$) to calculate $(x + y)$

x = number of electrons having $(\ell \times m) = -1$

y = number of electrons having $(\ell \times m) = -4$

then value of $(x + y)$ is equal to?

- (A) Number of p electrons in Al
- (B) Number of s electron in K
- (C) Number of d electron in Co
- (D) Period number of $_{89}\text{Ac}$

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
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Paragraph for Question no. 15 and 16

According to VBT the extent of overlapping depends upon types of orbitals involved in overlapping and nature of overlapping. More will be the overlapping than bond energy will also be high.

15. The **INCORRECT** order of bond strength will be :
- (A) $\text{H-H} > \text{Cl-Cl} > \text{Br-Br}$ (B) $\text{Cl-Cl} > \text{P-P} > \text{Si-Si}$
(C) $\text{H-Cl} > \text{H-Br} > \text{H-I}$ (D) $\text{F-F} > \text{Cl-Cl} > \text{Br-Br}$
16. If inter nuclear axis is x axis then which of the following combination of orbitals does not form covalent bond.
- (A) $s + p_y$ (B) $p_y + p_y$ (C) $d_{yz} + d_{yz}$ (D) $d_{xy} + d_{xy}$

Space for Rough Work

Paragraph for Question no. 17 and 18

Elements with their electronic configurations are given below :

I : $1s^2 2s^2$

II : $1s^2 2s^2 2p^6$

III : $1s^2 2s^2 2p^6 3s^2$

IV : $1s^2 2s^2 2p^3$

V : $1s^2 2s^2 2p^5$

17. Which of the following element having highest ionisation energy?

(A) I

(B) II

(C) III

(D) V

18. Which of the following is the **CORRECT** order of increasing size?

(A) $I < III < IV$

(B) $I > IV > V$

(C) $II < V < IV$

(D) $I > III > II$

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 21)**

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-
1. Minimum positive integral value of x satisfying inequality $2x^4 > 3x^3 + 9x^2$
(A) 3 (B) 4 (C) 0 (D) 10
 2. A group of members know at least one of the two languages, Hindi or Kannada. In the group, 150 members know Hindi and 80 members know Kannada and 70 of them know both. How many members are there in the group ?
(A) 230 (B) 160 (C) 150 (D) 170
 3. If $x > 1$, $y > 1$, $z > 1$ are in G.P., then $\log_{ex} e$, $\log_{ey} e$, $\log_{ez} e$ are in (where x, y, z are not equal)
(A) A.P. (B) H.P. (C) G.P. (D) A.G.P.
-

Space for Rough Work

4. Number of real values of x satisfying $5 - x^2 = \sqrt{5 - x}$ is
 (A) 0 (B) 4 (C) 2 (D) 1
5. $4(\sin^6\theta + \cos^6\theta) - 6(\sin^4\theta + \cos^4\theta)$ is equal to
 (A) 0 (B) 1 (C) -2 (D) none of these
6. The turning point (vertex) of the parabola $y = x^2 - 2ax + 1$ is closest to the origin when
 (A) $a = 0$ (B) $a = \pm 1$ (C) $a = \pm \frac{1}{\sqrt{2}}$ (D) $a = \pm \frac{1}{\sqrt{2}}$ or $a = 0$
7. If 1, p , q and 4 are in harmonic progression, then the value of $(3p + q)$ is equal to
 (A) 3 (B) 4 (C) 5 (D) 6

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

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8. If $2x - 3y - z = 0$ and $x + 3y - 14z = 0$, then value of $\frac{x^2 + 3xy}{y^2 + z^2}$ is
- (A) prime number (B) composite number
 (C) integer (D) irrational number
9. The value of the expression $\log_{10}(\tan 6^\circ) + \log_{10}(\tan 12^\circ) + \log_{10}(\tan 18^\circ) + \dots + \log_{10}(\tan 84^\circ)$ is
- (A) a whole number (B) an irrational number
 (C) a negative integer (D) a rational number which is an integer

Space for Rough Work

10. If the roots of $x^2 + ax + b = 0$ are $\sec^2 \frac{\pi}{8}$ and $\operatorname{cosec}^2 \frac{\pi}{8}$, then which of the following is correct?
 (A) $a - b = 0$ (B) $a + b = 0$ (C) $a^3 + b^3 = 0$ (D) $a = 2b$
11. If $\sum_{r=0}^{n-1} \log_2 \left(\frac{r+2}{r+1} \right) = \prod_{r=10}^{99} \log_r (r+1)$, then 'n' is equal to
 (A) 4 (B) 3 (C) $\tan^2 \frac{\pi}{3}$ (D) 6
12. Square root of the root of the equation $3x^{\log_3 4} + 4^{\log_3 x} = 64$ is equal to
 (A) 4 (B) 3 (C) $\tan^2 \frac{\pi}{3}$ (D) 6
13. The value of the expression $\log_{\left(\operatorname{cosec} \frac{\pi}{6}\right)^9} \left(\sqrt{3+2\sqrt{2}} + \sqrt{3-2\sqrt{2}} \right)$ can be expressed in the lowest form as $\frac{p}{q}$ where p and q are relatively prime, then answer the following
 (A) $p = 1$ (B) $q = 6$ (C) $p = 2$ (D) $q = 5$
14. The value of x satisfying the equation $\ln (\tan x) = 0$, is/are
 (A) $-\frac{7\pi}{4}$ (B) $-\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{\pi}{4}$

Space for Rough Work

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 - Based on each paragraph, there are **TWO** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
 - For each question, darken the bubble corresponding to the correct option in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.
-

Paragraph for Question no. 15 and 16

Let $f(x) = ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$. It is known that $f(5) = -3f(2)$ and that 3 is a root of $f(x) = 0$.

15. The other root of $f(x) = 0$ is
(A) -4 (B) 2 (C) 6 (D) can not be determined
16. The value of the sum $(a + b + c)$ is
(A) 9 (B) 14
(C) 13 (D) can not be uniquely determined
-

Space for Rough Work

Paragraph for Question no. 17 and 18

Let $S_r = r^2 + r + 1$, $S_r - S_{r-1} = Q_r$, $V_r = Q_r \cdot Q_{r+1} - 2$, $r \in W$.

17. Remainder obtained, when V_r is divided by 8 is-

- (A) 2 (B) 4 (C) 6 (D) 0

18. $\prod_{r=1}^n \frac{S_r}{S_{r-1}}$ is equal to-

- (A) S_n (B) S_{n-1} (C) 1 (D) V_n

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has one section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	7	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	21
I(ii)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	28
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator